

Chassis Shape Menu

What Shape Should Our Robot Base Have?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different frame shapes and imagine how each one could support the robot's job.
- **[Red: Critical Thinking & Problem Solving]** We compare trade-offs before building.
- **[Blue: Effective Communication]** We explain our design choice in the engineering portfolio.
- **[Green: Personal Development]** We make responsible choices based on time, skill, tools, and materials.

Decision Being Made

Your team is choosing the basic shape of the robot chassis. The chassis is the base of the robot. It holds the wheels, motors, battery, electronics, and future mechanisms such as a shooter, hopper, trigger, or intake.

A chassis is not just "the bottom of the robot." Its shape affects:

- how the robot drives,
- how stable the robot is,
- how much space is available inside,
- where balls or artifacts can move,
- where motors and electronics can go,
- how easy the robot is to repair,
- and how easy it will be to add mechanisms later.

Background Sources

Use ideas from:

- chassis design discussion,
- wheel and drive system notes,
- borrowed materials expectations,
- FIRST / FTC values,
- your team's strategy ideas,
- and any robot driving observations from Day 1.

Why This Decision Matters

Teams often want to start building immediately, but chassis shape is one of the most important early robot decisions. A strong, simple frame may drive well but leave less room for game pieces. An open frame may support an intake or pushing strategy but may need extra support to stay rigid. A narrow frame may park or share space more easily but may leave less room for electronics and mechanisms.

There is no perfect shape. Your job is to choose a shape that is good enough to build, test, and improve.

How to Use This Menu

- Read the chassis shape options.
- Discuss what each shape helps and what it makes harder.
- Think about where wheels, motors, battery, electronics, and future mechanisms might go.
- Choose a first chassis shape.
- Sketch your shape from the top view.
- Record your choice, reasoning, trade-off, and evidence plan in the portfolio.

Menu Options

Option A: O-Shape / Box Frame

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The frame is closed on all four sides, like a rectangle or box. It may look like an "O" from the top.

What it helps: This is often strong, simple, and easier to square. It gives the robot a clear outside frame and can protect internal parts.

Best for: Teams that want a reliable driving base, simple construction, and clear places to mount motors, wheels, battery, and electronics.

Ease of use: Easy

Likelihood of success: High

Time to implement: Fast to moderate

Material needs: Usually needs enough channel or extrusion to make a full outside frame. It may require corner brackets, plates, or inside connectors.

Possible trade-offs: A closed front may block balls, artifacts, or intake paths. It may be harder to push game pieces through the robot. Future mechanisms may need to be mounted above or outside the frame.

Evidence we might watch: Does the frame stay square? Does it drive straight? Is there enough room inside for battery, electronics, motors, and mechanisms?

Option B: H-Shape Frame

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The frame has two side rails and a cross member in the middle, like the letter "H."

What it helps: An H-shape can leave open space at the front and back. This may help with intakes, pushing balls, or interacting with game pieces from either side.

Best for: Teams that want openings at both ends of the robot or want to leave the front and back flexible for future mechanisms.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Material needs: Needs two strong side rails and at least one cross member. The team may need brackets or plates to keep the frame rigid.

Possible trade-offs: The frame may twist more than a closed box frame. The middle cross member may block the path for a ball, hopper, battery, or electronics. The team must decide what the open spaces are for.

Evidence we might watch: Does the frame twist during turns? Can balls or artifacts move through the open spaces? Does the cross member block anything important?

Option C: A-Shape / One-End-Open Frame

Graduate Expectation Connection: [Purple: Curiosity & Creativity]

What it means: The frame is closed on one end and open on the other. It is like an H-shape where one end is closed. In this course, "A-shape" means one end is open for interaction with game pieces; it does not need to be angled.

What it helps: The open end can guide balls or artifacts, make space for an intake, or allow the robot to push game pieces toward the human operator or scoring area. The closed end can add strength and give the team a stable place for battery, electronics, or a shooter mount.

Best for: Teams that want one clear "front" of the robot for collecting, pushing, or interacting with game pieces.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Material needs: Needs side rails, a rear cross member, and possibly extra brackets or plates to reduce twisting at the open end.

Possible trade-offs: The open end may be less rigid. The team must clearly define the robot's front and protect wheels, wires, and mechanisms near the opening.

Evidence we might watch: Does the open end help with balls or artifacts? Does the frame twist? Can the team still mount wheels, motors, electronics, and battery safely?

Option D: Narrow Frame

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The robot is intentionally built narrower than the maximum allowed size.

What it helps: A narrow robot may fit through tighter spaces, share the field with other robots, park more easily, or move around crowded areas.

Best for: Teams that value maneuverability, parking, passing, or working near other robots.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Material needs: May use shorter channels or extrusions, but internal space becomes limited.

Possible trade-offs: Less room for battery, electronics, shooter, hopper, trigger, or intake. The robot may be less stable if it becomes tall or top-heavy.

Evidence we might watch: Can the robot still fit all required parts? Does it tip during turns? Does the narrow shape improve parking or movement?

Option E: Wide / Full-Size Frame

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The robot uses most of the allowed footprint.

What it helps: A wider frame gives more room for mechanisms, electronics, battery, and stable mounting. It may be easier to build and more stable.

Best for: Teams that need room for mechanisms and want a stable base.

Ease of use: Easy to medium

Likelihood of success: High

Time to implement: Moderate

Material needs: May require longer channels or extrusions. Long pieces may be limited.

Possible trade-offs: A wide robot may have trouble sharing space, parking with other robots, or moving through tight areas.

Evidence we might watch: Does the extra space help the build? Does the size create problems with parking, turning, or crowding?

Shape Comparison Table

Shape	Best Feature	Main Risk	Good For
O-shape / box	Strong and simple	May block game-piece paths	Reliable driving base
H-shape	Open front and back	May twist or block middle path	Flexible intake/pushing options
A-shape / one-end-open	One clear open end	Open end may need support	Front intake, pushing, or guiding
Narrow frame	Easier to fit in tight spaces	Less room inside	Parking, passing, shared field space
Wide / full-size frame	Stable and roomy	Harder to share space	Mechanisms and stability

Materials Note

Your team may use parts from different systems, including **goBILDA**, **REV**, and **Tetrix**. These systems may use different hole patterns, sizes, connectors, and fasteners.

You may have access to:

- U-channel from goBILDA, REV, or Tetrix,
- REV 15 mm extrusion,
- brackets,
- plates,

- inside connectors,
- bolts, nuts, and spacers.

You will not have unlimited access to every length or every part. Your chassis shape must work with the materials that are actually available.

This is part of engineering. Real teams design under constraints.

Chassis Shape Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Chassis shape				
Robot width				
Open spaces				
Cross members / support				

Sketch Requirement

Draw a top-view sketch of your chassis plan.

Your sketch should label:

- front of robot,
- back of robot,
- frame shape,
- drive wheels,
- omni wheels,
- motor locations,
- battery location,
- electronics location,
- open space for balls, artifacts, intake, shooter, hopper, or trigger.

Must / Should / Could

Level	Expectation
Must Do	Choose a chassis shape and draw a labeled top-view sketch.
Should Do	Explain why the shape fits your team's strategy and name at least one trade-off.

Level	Expectation
Could Do	Compare two chassis shapes, explain how each would affect scoring or future mechanisms, and choose using evidence or measurements.

SLD Connection

Main SLD Prompt

Should our chassis be simple and strong, or open and flexible for future mechanisms?

Supporting Questions

- What does our robot need to do first?
- Where might balls or artifacts need to move?
- Where will the battery and electronics go?
- What shape gives us the best chance of building a working robot quickly?
- What shape leaves room for future mechanisms?
- What material constraints might change our first choice?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected chassis shape,
- labeled top-view sketch,
- explanation of why the team chose the shape,
- trade-off statement,
- materials the team expects to use,
- evidence the team will watch during testing.

Reflection

Our chosen chassis shape is:

We chose this shape because:

The main trade-off we accepted is:

The evidence we will watch during testing is:
